# A MORPHISM OF TRIANGLES WHOSE FIRST TWO COMPONENTS ARE HOMOTOPY EQUIVALENCES HAS THE THIRD ONE TOO

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*Editorial note.* The counted statement is the author's own lemma environment labelled lemma-third-isomorphism (source lines 1015-1028); the proof environment begins two lines below it and is its own proof, closing at source line 1062.

*Editorial note.* The proof invokes two author results of the same chapter by label, namely the functoriality of the cone and the nilpotence of a composite of maps of admissible sequences; both are retained as context spans, and the generator expands that context transitively, so every reference resolves inside the assembled package.

*Editorial note.* The proof displays one Xy-pic diagram of three rows; it is part of the retained source range and is reproduced as written, so no figure file is needed.

*Editorial note.* The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

*Editorial scope.* One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: in a morphism of triangles of dg modules, if the first two maps are homotopy equivalences then so is the third.

*Allowed standard prerequisites (editorial).*

- Cone of a morphism and its triangle. The cone C(f) of a map of dg modules with its inclusion, projection and the triangle it forms.. The triangles and their cones are the objects of the morphism (a, b, c) (source lines 1019-1026).
- Homotopy equivalence and inverse in the homotopy category. A map is a homotopy equivalence when it has an inverse in the homotopy category of dg modules.. The hypothesis gives inverses a^{-1} and b^{-1}, and the conclusion is that c has one (source lines 1027, 1031-1034, 1060-1061).
- Functoriality of the cone and nilpotence of a composite. A homotopy commutative square induces a map of the cones, and a composite of two compatible maps of admissible sequences is homotopic to zero (the author's lemmas lemma-functorial-cone and lemma-nilpotent, retained as context).. The first builds c'; the second shows that the composites of c and c' have square zero, which is what makes 2 - c an inverse (source lines 1035, 1056-1061).

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*Date*: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $(A, d)$ be a differential graded algebra. Suppose that*

$$\begin{array}{ccc} K_1 & \xrightarrow[f_1]{} & L_1 \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ K_2 & \xrightarrow{f_2} & L_2 \end{array}$$

*is a diagram of homomorphisms of differential graded $A$-modules which is commutative up to homotopy. Then there exists a morphism $c : C(f_1) \to C(f_2)$ which gives rise to a morphism of triangles*

$$(a, b, c) : (K_1, L_1, C(f_1), f_1, i_1, p_1) \to (K_1, L_1, C(f_1), f_2, i_2, p_2)$$

*in $K(Mod_{(A,d)})$.*

*Proof.* Let $h : K_1 \to L_2$ be a homotopy between $f_2 \circ a$ and $b \circ f_1$. Define $c$ by the matrix

$$c = \begin{pmatrix} b & h \\ 0 & a \end{pmatrix} : L_1 \oplus K_1 \to L_2 \oplus K_2$$

A matrix computation show that $c$ is a morphism of differential graded modules. It is trivial that $c \circ i_1 = i_2 \circ b$, and it is trivial also to check that $p_2 \circ c = a \circ p_1$. □

**Lemma 0.2.** *Let $(A, d)$ be a differential graded algebra. Let $0 \to K_i \to L_i \to M_i \to 0$, $i = 1, 2, 3$ be admissible short exact sequence of differential graded $A$-modules. Let $b : L_1 \to L_2$ and $b' : L_2 \to L_3$ be homomorphisms of differential graded modules such that*

$$\begin{array}{ccccc} K_1 & \longrightarrow & L_1 & \longrightarrow & M_1 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b}\downarrow & & {\scriptstyle 0}\downarrow \\ K_2 & \longrightarrow & L_2 & \longrightarrow & M_2 \end{array} \quad \text{and} \quad \begin{array}{ccccc} K_2 & \longrightarrow & L_2 & \longrightarrow & M_2 \\ {\scriptstyle 0}\downarrow & & {\scriptstyle b'}\downarrow & & {\scriptstyle 0}\downarrow \\ K_3 & \longrightarrow & L_3 & \longrightarrow & M_3 \end{array}$$

*commute up to homotopy. Then $b' \circ b$ is homotopic to $0$.*

*Proof.* By Lemma 0.3 we can replace $b$ and $b'$ by homotopic maps such that the right square of the left diagram commutes and the left square of the right diagram commutes. In other words, we have $\mathrm{Im}(b) \subset \mathrm{Im}(K_2 \to L_2)$ and $\mathrm{Ker}(b') \supset \mathrm{Im}(K_2 \to L_2)$. Then $b' \circ b = 0$ as a map of modules. □

**Lemma 0.3.** *Let $(A, d)$ be a differential graded algebra. Let*

$$\begin{array}{ccc} K & \xrightarrow[f]{} & L \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ M & \xrightarrow{g} & N \end{array}$$

*be a diagram of homomorphisms of differential graded $A$-modules commuting up to homotopy.*

1. *If $f$ is an admissible monomorphism, then $b$ is homotopic to a homomorphism which makes the diagram commute.*
2. *If $g$ is an admissible epimorphism, then $a$ is homotopic to a morphism which makes the diagram commute.*

*Proof.* Let $h : K \to N$ be a homotopy between $bf$ and $ga$, i.e., $bf - ga = \mathrm{d}h + h\mathrm{d}$. Suppose that $\pi : L \to K$ is a graded $A$-module map left inverse to $f$. Take $b' = b - \mathrm{d}h\pi - h\pi\mathrm{d}$. Suppose $s : N \to M$ is a graded $A$-module map right inverse to $g$. Take $a' = a + \mathrm{d}sh + sh\mathrm{d}$. Computations omitted. □

**Lemma 0.4.** *Let $(A, d)$ be a differential graded algebra. Let $f_1 : K_1 \to L_1$ and $f_2 : K_2 \to L_2$ be homomorphisms of differential graded $A$-modules. Let*

$$(a, b, c) : (K_1, L_1, C(f_1), f_1, i_1, p_1) \longrightarrow (K_1, L_1, C(f_1), f_2, i_2, p_2)$$

*be any morphism of triangles of $K(Mod_{(A,d)})$. If $a$ and $b$ are homotopy equivalences then so is $c$.*

*Proof.* Let $a^{-1} : K_2 \to K_1$ be a homomorphism of differential graded $A$-modules which is inverse to $a$ in $K(\mathrm{Mod}_{(A,\mathrm{d})})$. Let $b^{-1} : L_2 \to L_1$ be a homomorphism of differential graded $A$-modules which is inverse to $b$ in $K(\mathrm{Mod}_{(A,\mathrm{d})})$. Let $c' : C(f_2) \to C(f_1)$ be the morphism from Lemma 0.1 applied to $f_1 \circ a^{-1} = b^{-1} \circ f_2$. If we can show that $c \circ c'$ and $c' \circ c$ are isomorphisms in $K(\mathrm{Mod}_{(A,\mathrm{d})})$ then we win. Hence it suffices to prove the following: Given a morphism of triangles $(1, 1, c) : (K, L, C(f), f, i, p)$ in $K(\mathrm{Mod}_{(A,\mathrm{d})})$ the morphism $c$ is an isomorphism in $K(\mathrm{Mod}_{(A,\mathrm{d})})$. By assumption the two squares in the diagram

$$\begin{array}{ccccc} L & \longrightarrow & C(f) & \longrightarrow & K[1] \\ {\scriptstyle 1}\downarrow & & {\scriptstyle c}\downarrow & & {\scriptstyle 1}\downarrow \\ L & \longrightarrow & C(f) & \longrightarrow & K[1] \end{array}$$

commute up to homotopy. By construction of $C(f)$ the rows form admissible short exact sequences. Thus we see that $(c - 1)^2 = 0$ in $K(\mathrm{Mod}_{(A,\mathrm{d})})$ by Lemma 0.2. Hence $c$ is an isomorphism in $K(\mathrm{Mod}_{(A,\mathrm{d})})$ with inverse $2 - c$. □